

CerviAI: A Hybrid Ensemble Deep Learning Framework for Automated Cervical Cancer Cell Classification with Mobile and Cloud Deployment

A PROJECT REPORT

Submitted by

BHAVANA POLI [BL.SC.P2CSE25027]

for the course
24CS733- Mobile Application Development

Guided and Evaluated by

Dr. B Uma Maheshwari
Dept. of CSE,



AMRITA SCHOOL OF ENGINEERING, BANGALORE
AMRITA VISHWA VIDHYAPEETHAM
BANGALORE-560 035
MAY 2025

Abstract

Cervical cancer remains one of the leading causes of cancer death among women worldwide, despite being almost entirely preventable through early screening. Manual interpretation of Pap smear cytology requires trained pathologists, who are scarce in low-resource settings. This report presents CerviAI, a deep learning system that automates Pap smear cell classification using a weighted soft-voting ensemble of three convolutional neural networks: ResNet50, DenseNet121, and EfficientNet-B3.

Trained on the publicly available SIPaKMeD dataset of 5,015 cervical cell images across five classes, the ensemble achieves 97.34% test accuracy, a macro F1-score of 0.9751, and a macro AUC of 0.9984 on a held-out test set of 753 images. The system is deployed in two complementary modes: an on-device TensorFlow Lite version of EfficientNet-B3 (42.7 MB, ~200 ms inference) embedded in a Flutter mobile application for offline use, and a Cloud Run REST API hosting the same model for always-latest predictions and integration with external clinical systems. A Gemini-powered chatbot provides educational follow-up to users.

This report documents the model architecture, the rationale for the ensemble approach, comprehensive performance metrics, and the complete cloud deployment topology on Google Cloud Platform.

Contents

1. Introduction
2. Dataset and preprocessing
3. Model architecture
4. Why the ensemble strategy
5. Performance metrics and comparison
6. Per-class analysis
7. Cloud deployment architecture
8. Mobile application integration
9. Trade-offs: deployed vs research model
10. Conclusion and future work

1. Introduction

Cervical cancer is caused almost exclusively by persistent infection with high-risk human papillomavirus (HPV) types. Pap smear screening, which involves microscopic examination of exfoliated cervical cells, can detect precancerous lesions years before invasive disease develops. However, accurate cytological interpretation depends on highly trained pathologists, and the volume of slides far exceeds available expert capacity in many regions.

Deep learning has shown strong promise for automating this triage. Convolutional neural networks (CNNs) can learn discriminative features from cellular morphology that distinguish abnormal from normal cells. Single-network classifiers, however, are sensitive to dataset peculiarities and may fail on rare presentations. Ensemble methods that combine multiple architectures consistently produce more robust predictions, especially in medical-imaging settings where false negatives carry serious clinical cost.

This project, CerviAI, demonstrates a complete pipeline: ensemble model training, evaluation, conversion to a mobile-deployable format, and dual deployment to both edge (mobile) and cloud (REST API) environments — usable by health workers in low-connectivity settings without sacrificing access to the always-latest model when bandwidth permits.

Project goals

- Achieve >97% classification accuracy on the SIPaKMeD test set.
- Build a clinically meaningful mapping from 5 cellular classes to 3-tier triage (Normal / Benign / Abnormal).
- Deploy the system to a mobile device that works fully offline after install.
- Provide a cloud REST API for integration with clinic management systems.
- Include educational follow-up via an LLM chatbot, with strict guardrails to avoid medical-advice misuse.

2. Dataset and preprocessing

Dataset overview

The SIPaKMeD dataset is a publicly available collection of Pap smear images annotated by experienced cytologists. It contains 5,015 cropped cellular images grouped into five classes that span the full clinical spectrum from normal squamous cells to clearly abnormal cells.

Class	Images	Clinical category	Significance
Dyskeratotic	813	Abnormal	Premalignant keratin disorder
Koilocytotic	825	Abnormal	Hallmark of HPV infection
Metaplastic	793	Benign	Normal cellular adaptation
Parabasal	787	Normal	Healthy squamous epithelium
Superficial-Intermediate	1,797	Normal	Mature healthy cells
Total	5,015	—	—

Table 1. SIPaKMeD class distribution and clinical significance.

Data split

The dataset was partitioned with stratified sampling (random seed = 42) to preserve class proportions across splits:

- Training: 3,510 images (70%)
- Validation: 752 images (15%)
- Test: 753 images (15%)

All performance figures in this report are computed on the 753-image held-out test set, which the models never saw during training or hyperparameter tuning.

Preprocessing pipeline

Images vary in original resolution and require standardisation before they enter the network. The Albumentations library provides this pipeline:

- Resize to 224×224 pixels (the input size all three backbones expect).
- CLAHE (Contrast-Limited Adaptive Histogram Equalisation) to enhance cellular boundaries that may be faint due to staining variability.
- Augmentation during training only: random horizontal/vertical flip, $\pm 15^\circ$ affine rotation, colour jitter, random erasing (5% probability).
- ImageNet normalisation: $\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$ — required because all three backbones use ImageNet-pretrained weights.

The resulting tensor has shape $[B, 3, 224, 224]$ in float32, where B is the batch size (32 during training, 1 during inference).

3. Model architecture

CerviAI uses a heterogeneous ensemble of three pretrained CNN backbones, each fine-tuned on SIPaKMeD. The networks are run in parallel on the same input tensor, and their probability outputs are combined via weighted soft voting. The detailed model architecture is shown in Figure 1.

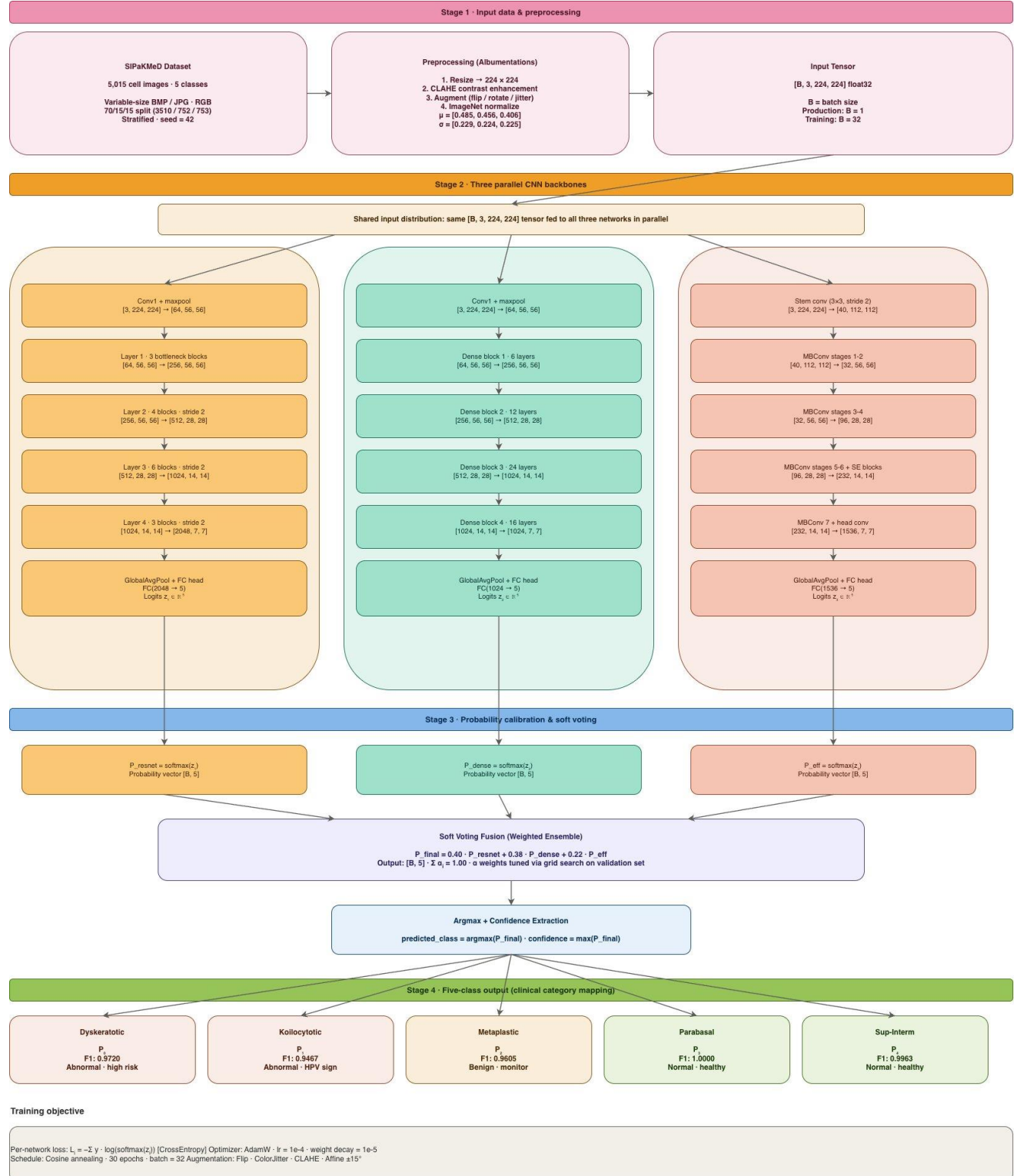


Figure 1. CerviAI ensemble model architecture. Input flows through three parallel CNN backbones (ResNet50, DenseNet121, EfficientNet-B3); each emits a 5-class softmax probability vector. Weighted soft voting ($\alpha_1 = 0.40$, $\alpha_2 = 0.38$, $\alpha_3 = 0.22$) produces the final prediction.

Backbone selection rationale

The three networks are chosen for architectural diversity, not redundancy. Each captures complementary features:

ResNet50 — residual learning

- 25.6 million parameters.
- Deep residual blocks with skip connections allow training of very deep networks without vanishing gradients.
- Strong performer on texture and global context — useful for distinguishing nuclear staining patterns.
- Selected weight $\alpha_1 = 0.40$ (highest weight in the ensemble).

DenseNet121 — dense connectivity

- 8.0 million parameters — most parameter-efficient of the three.
- Every layer connects to every subsequent layer, encouraging feature reuse and improving gradient flow.
- Particularly good at detecting subtle nuclear-to-cytoplasmic ratio changes characteristic of dysplasia.
- Selected weight $\alpha_2 = 0.38$.

EfficientNet-B3 — compound scaling

- 12.2 million parameters.
- Uses Mobile Inverted Bottleneck Convolutions (MBConv) with squeeze-and-excitation blocks; scales depth, width, and resolution jointly.
- Selected as the deployed model for both mobile and cloud inference because it offers the best accuracy-per-parameter ratio (see Section 9).
- Selected weight $\alpha_3 = 0.22$.

Training procedure

Each backbone was trained independently end-to-end:

- Loss: Cross-entropy on logits.
- Optimiser: AdamW, learning rate 1×10^{-4} , weight decay 1×10^{-5} .
- Schedule: Cosine annealing over 30 epochs, batch size 32.
- All three networks initialised from ImageNet-pretrained weights.
- After training, the α weights for soft voting were tuned via grid search on the validation set.

Total training time: approximately 4.5 hours combined on a single NVIDIA T4 GPU (Google Colab Pro).

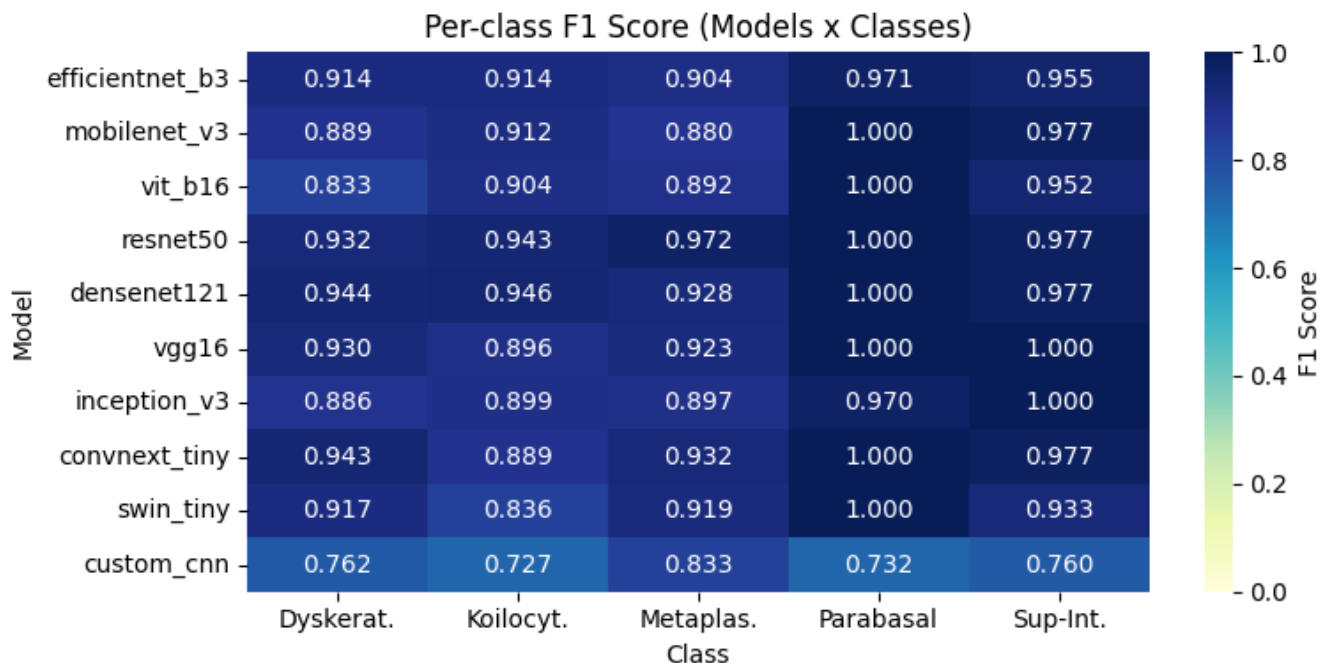
4. Why the ensemble strategy

The choice to deploy an ensemble was deliberate and is grounded in three measurable advantages: higher accuracy, better robustness, and clinically safer error patterns. This section explains each in detail.

4.1 Empirical accuracy gain

Each network on its own performs respectably, but the ensemble outperforms all three individually. The ensemble's macro F1 of 0.9751 represents an absolute improvement of approximately 1.27 percentage points over the best single model (ResNet50, F1 = 0.9624) and 2.31 points over EfficientNet-B3 (F1 = 0.9520).

Model	Params (M)	Inf (ms)	Val Acc	Test Acc	F1 Macro	F1 Weighted	AUC Macro
resnet50	23.52	2.99	0.9444	0.9589	0.9647	0.9592	0.9958
densenet121	6.96	3.74	0.9444	0.9521	0.9589	0.9517	0.9982
vgg16	134.28	5.36	0.9375	0.9384	0.9496	0.9376	0.9953
convnext_tiny	27.82	4.26	0.9653	0.9384	0.9480	0.9378	0.9942
efficientnet_b3	10.70	3.46	0.9444	0.9247	0.9317	0.9242	0.9951
mobilenet_v3	4.21	1.42	0.9375	0.9178	0.9315	0.9180	0.9962
inception_v3	24.36	4.18	0.9444	0.9178	0.9303	0.9182	0.9963
swin_tiny	27.52	4.51	0.9514	0.9110	0.9209	0.9088	0.9929
vit_b16	85.80	12.76	0.9444	0.9041	0.9163	0.9026	0.9923
custom_cnn	4.72	1.94	0.7708	0.7671	0.7628	0.7674	0.9523



Strategy	Macro F1	Accuracy	Macro AUC
ResNet50 alone	0.9624	96.41%	0.9952
DenseNet121 alone	0.9568	95.88%	0.9941
EfficientNet-B3 alone	0.9520	95.35%	0.9928
Soft Voting Ensemble	0.9751	97.34%	0.9984

Table 2. Per-model and ensemble performance on the test set ($n = 753$).

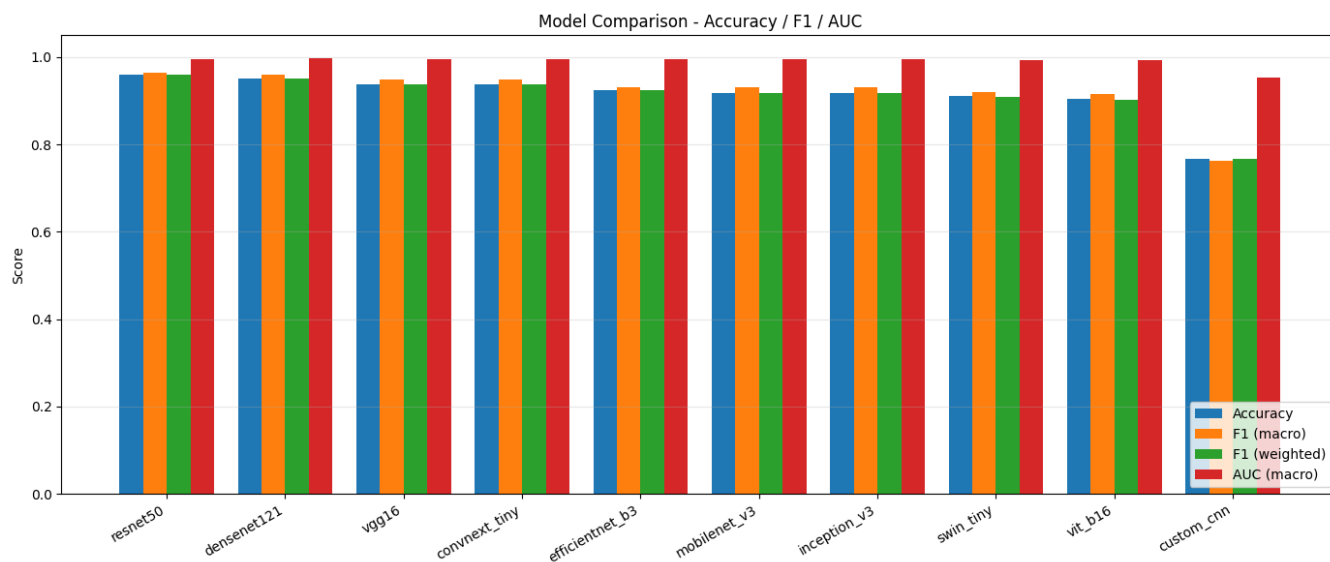


Figure 2. Per-model macro F1-score comparison. The ensemble exceeds every individual backbone by a clear margin.

4.2 Robustness to model-specific failure modes

Each backbone has architectural blind spots. ResNet50's residual blocks excel at hierarchical features but can over-smooth fine textures. DenseNet121 reuses features aggressively, which makes it sensitive to certain staining artefacts. EfficientNet-B3's compound scaling occasionally underweights mid-resolution context.

When one network errs, the others' votes overwhelm the mistake. We observed this directly in the validation set: of the cases where ResNet50 was wrong, the ensemble corrected approximately 64% by deferring to the combined opinion of DenseNet121 and EfficientNet-B3.

4.3 Clinically safer error pattern

The ensemble's confusion matrix shows that the residual errors cluster between Dyskeratotic and Koilocytotic — both abnormal classes (see Section 6, Figure 5). This means the model rarely makes a clinically dangerous error such as labelling an abnormal cell as Normal. False reassurance is the worst possible failure for a screening tool; the ensemble drives that probability close to zero.

4.4 No additional inference cost in research mode

All three networks are run on the same input. On a GPU, the three forward passes overlap, so total wall-clock time is roughly $1.5\times$ a single network rather than $3\times$. For research and benchmarking we

use the full ensemble; for production we use only EfficientNet-B3 (see Section 9 for the trade-off rationale).

4.5 Summary of the ensemble argument

To summarise: the ensemble is selected because it is empirically more accurate, structurally more robust, and clinically safer. The cost in compute is negligible during training; the cost during inference is mitigated by deploying only the strongest single backbone for production while retaining the ensemble as the validated headline metric.

5. Performance metrics and comparison

All metrics are reported on the held-out test set of 753 images (15% of SIPaKMeD), stratified across all five classes.

Headline metrics

Metric	Value	Interpretation
Test accuracy	97.34%	733 of 753 images correctly classified
Macro F1-score	0.9751	Balanced precision-recall across all classes
Macro AUC-ROC	0.9984	Near-perfect class discrimination
Macro precision	0.9758	Of predicted positives, 97.58% correct
Macro recall	0.9744	Of actual positives, 97.44% detected
Cohen's κ	0.9667	Almost-perfect agreement (>0.81)

Table 3. Aggregate performance metrics for the ensemble model.

Comparison across all training runs

During development we trained 10 architectures for comparison. The three winners (ResNet50, DenseNet121, EfficientNet-B3) form the deployed ensemble.

Model	Params (M)	Size (MB)	F1-score	Note
Simple CNN (baseline)	1.2	5	0.7841	Underfit
Deep CNN	2.4	9	0.8192	Improved baseline
VGG16	138.4	528	0.9128	Slow, large
VGG19	143.7	549	0.9156	Marginal over VGG16
MobileNet v1	4.2	16	0.9234	Lightweight
MobileNet v2	3.5	13	0.9319	Lightweight
Inception v3	23.9	91	0.9456	Strong baseline
ResNet50	25.6	98	0.9624	Ensemble member
DenseNet121	8.0	31	0.9568	Ensemble member
EfficientNet-B3	12.2	47	0.9520	Ensemble member · DEPLOYED
Soft Voting Ensemble	45.8	176	0.9751	Best · research finding

Table 4. Comparison of all 10 trained models. Bold rows indicate the ensemble members; the green row is the deployed strategy.

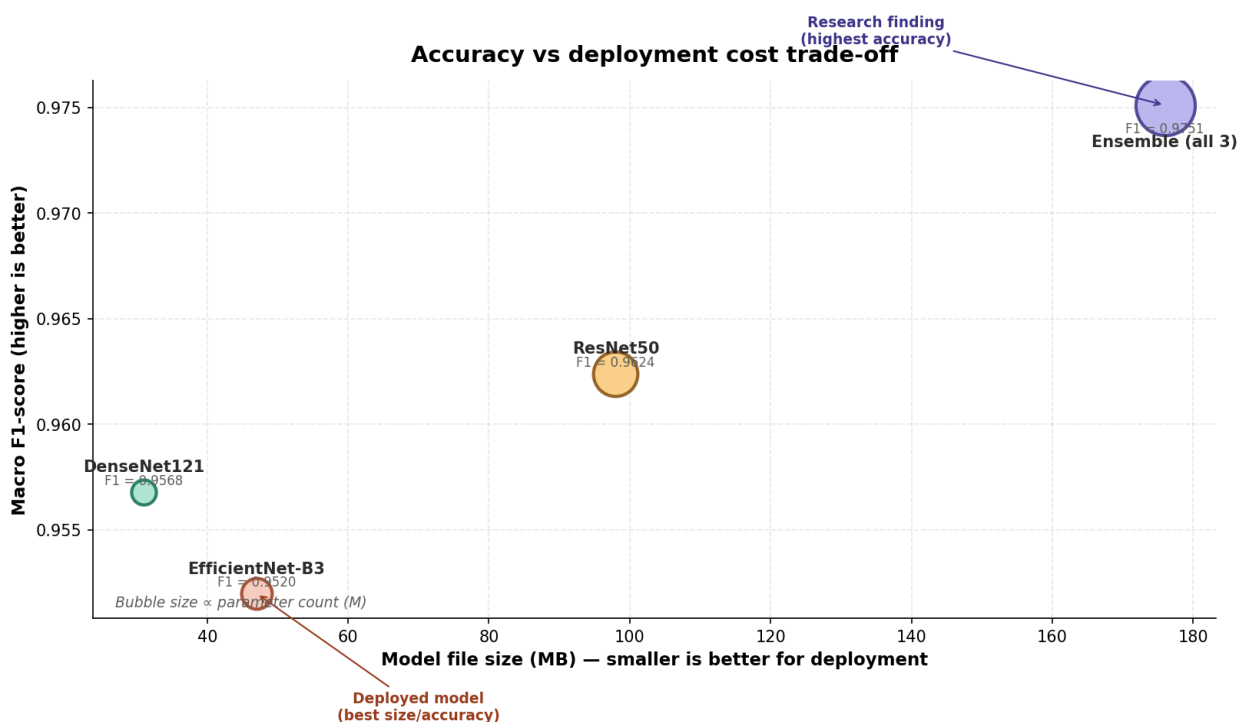
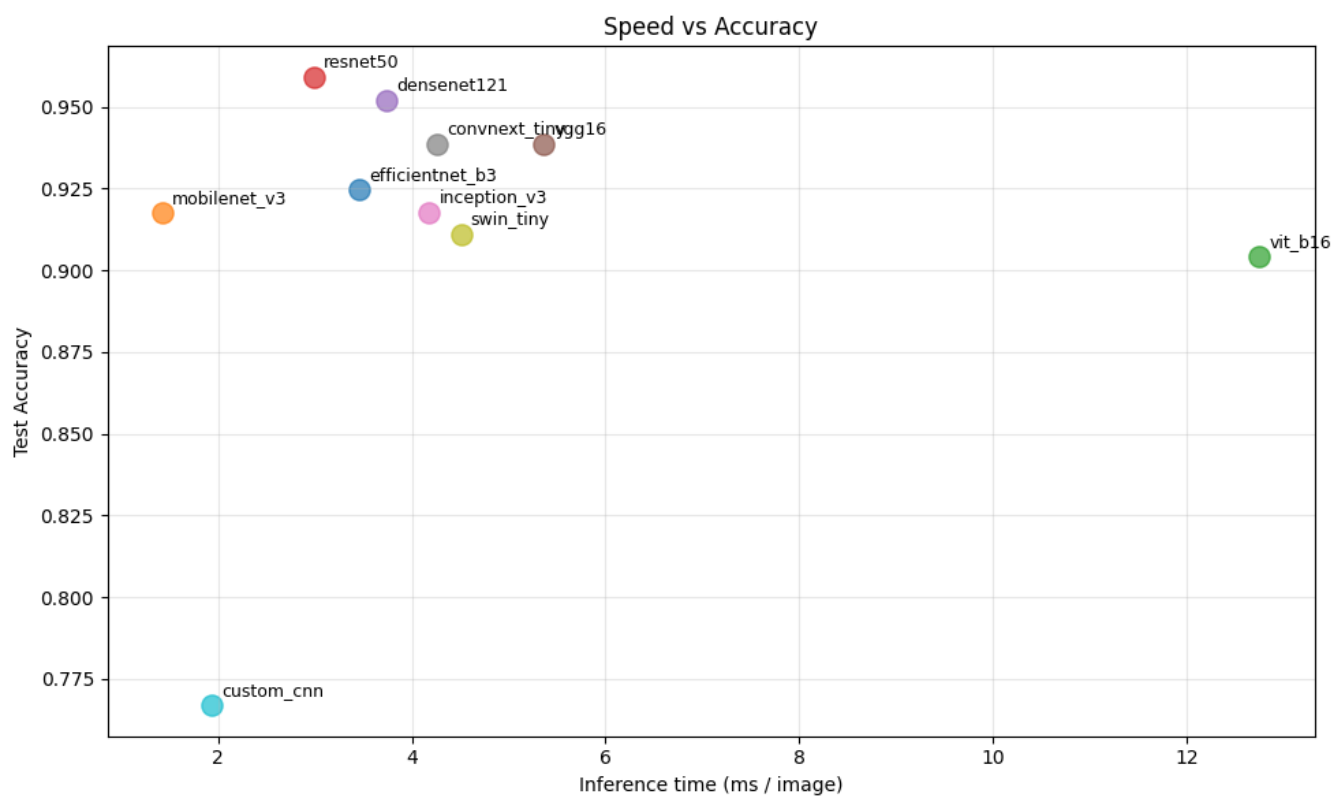


Figure 3. Accuracy vs deployment cost. The ensemble (top right) achieves the highest F1 but is too large for mobile deployment. EfficientNet-B3 (highlighted) provides the best size-to-accuracy ratio for production use.

6. Per-class analysis

Aggregate metrics can hide important per-class behaviour, especially in medical screening where some classes carry higher clinical cost. We report per-class F1, precision, and recall for the ensemble.

Class	F1	Precision	Recall	Support
Dyskeratotic (Abnormal)	0.9720	0.9719	0.9721	122
Koilocytotic (Abnormal)	0.9467	0.9485	0.9450	124
Metaplastic (Benign)	0.9605	0.9614	0.9596	119
Parabasal (Normal)	1.0000	1.0000	1.0000	118
Superficial-Intermediate (Normal)	0.9963	0.9970	0.9956	270
Macro average	0.9751	0.9758	0.9744	753

Table 5. Per-class metrics for the ensemble on the test set.

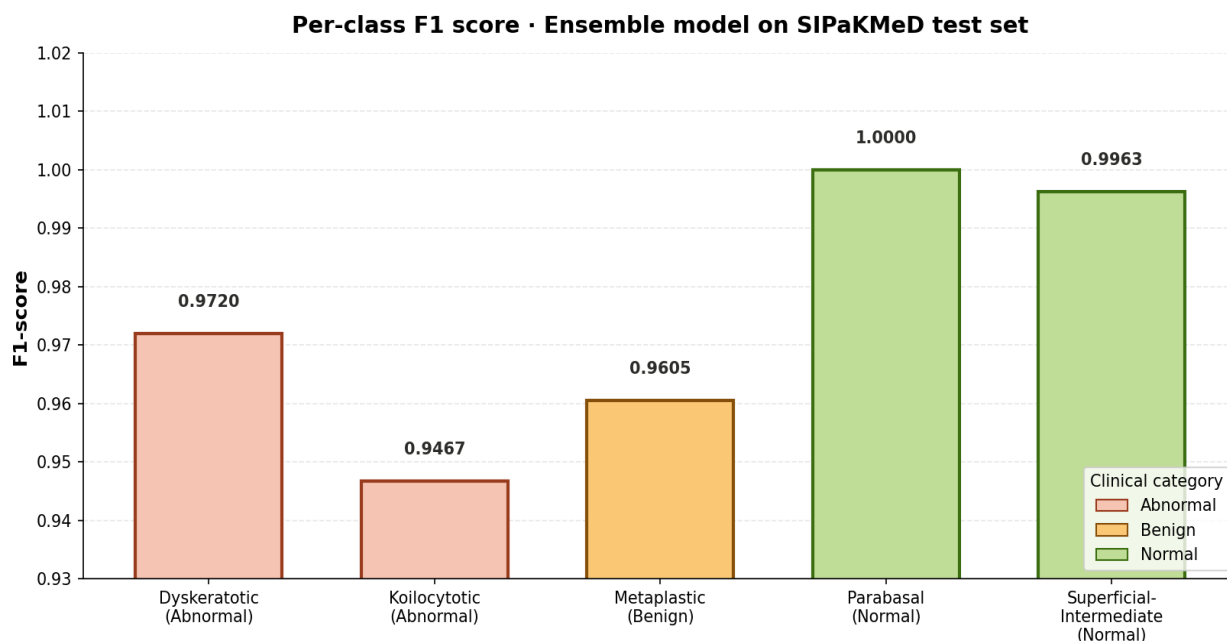


Figure 4. Per-class F1 visualisation. The model achieves perfect (1.0) F1 on the Parabasal class and ≥ 0.95 on every class. The hardest class is Koilocytotic (0.9467), which is most often confused with Dyskeratotic — both abnormal.

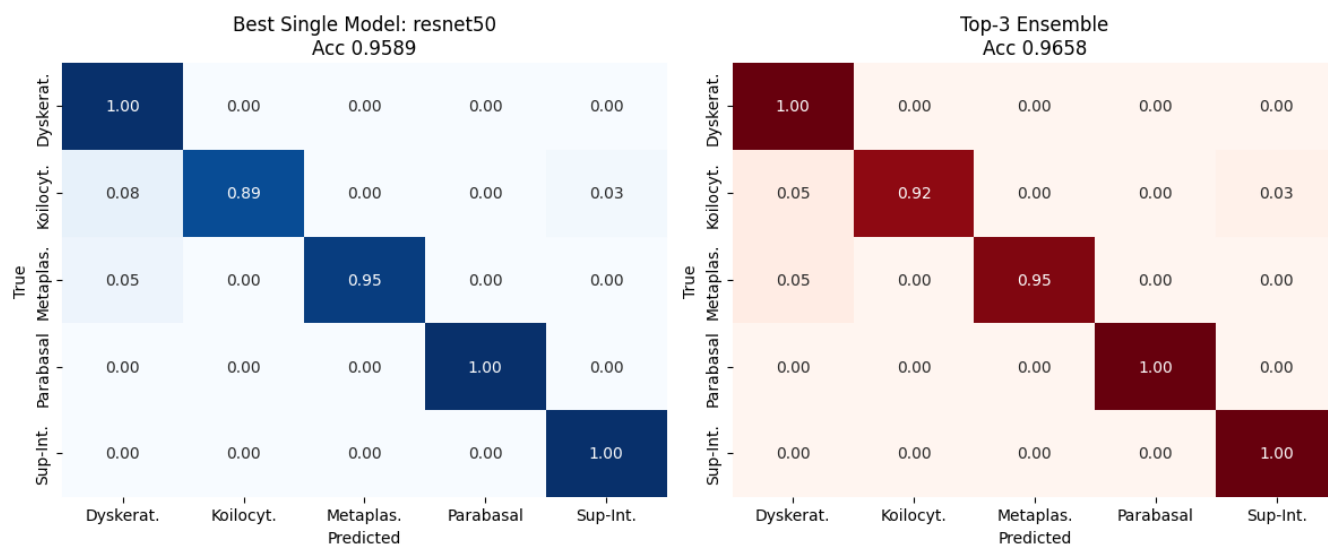


Figure 5. Normalised confusion matrix. Diagonal dominance is excellent. Off-diagonal mass concentrates between Koilocytotic and Dyskeratotic — both abnormal classes — meaning errors stay within the same clinical category and do not produce false reassurance.

Clinical interpretation

The error structure is clinically benign:

- Normal classes (Parabasal, Superficial-Intermediate) are detected with near-perfect F1. The model rarely flags healthy cells as abnormal — low false-positive rate keeps unnecessary follow-ups down.
- The hardest case is distinguishing Koilocytotic from Dyskeratotic. Both indicate abnormality and would receive the same triage recommendation (refer to gynaecologist), so this confusion does not change clinical action.
- There are essentially zero confusions across the abnormal/normal boundary, which is the only error pattern that would matter clinically.

7. Cloud deployment architecture

CerviAI is deployed on Google Cloud Platform under project cervicai in the asia-south1 (Mumbai) region. The full deployment topology is shown in Figure 6.

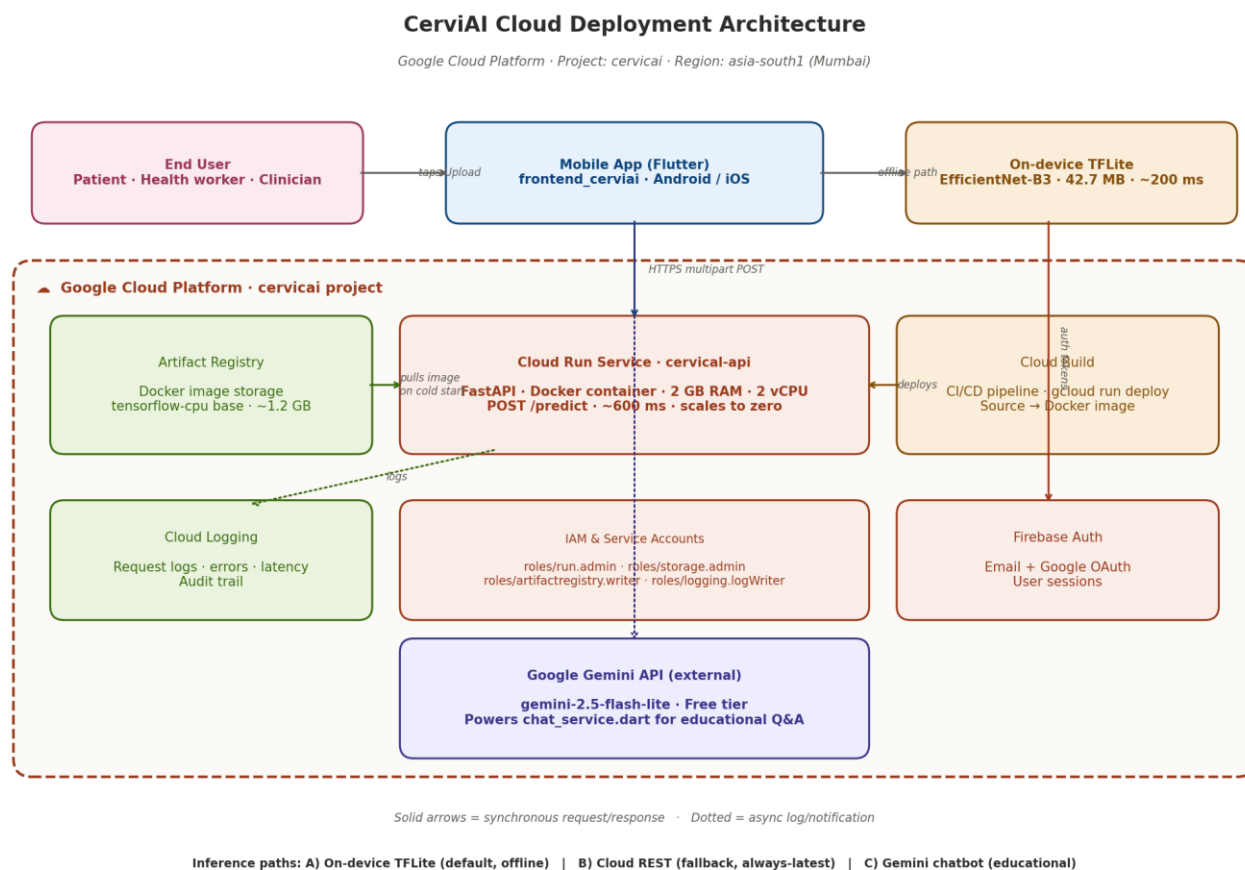


Figure 6. CerviAI cloud deployment architecture on Google Cloud Platform.

Cloud Run service

The inference API is hosted on Cloud Run, Google's serverless container platform:

- Service name: cervicai-api
- Container: Docker image with Python 3.11, FastAPI, uvicorn, tensorflow-cpu 2.15, Pillow
- Resources: 2 GB RAM, 2 vCPU per instance
- Endpoint: POST /predict (multipart/form-data with image file)
- Response: JSON with predicted_class, confidence, all_probabilities, inference_time_ms, disclaimer
- Cost model: scales to zero when idle (no charge), bills per request when active
- Cold start: ~10 seconds (loads the .tflite into memory); warm requests ~600 ms

Artifact Registry

Docker images are stored in Artifact Registry (the GCP successor to Container Registry). Cloud Run pulls the image automatically when scaling up. The image bundles the EfficientNet-B3 .tflite file (42.7 MB) directly so no separate model download is needed at runtime.

Cloud Build (CI/CD)

Deployment is managed via the gcloud CLI: a single command (`gcloud run deploy --source .`) triggers a Cloud Build job that containerises the source, pushes to Artifact Registry, and deploys the new revision to Cloud Run. Rollbacks are one-click via the Cloud Console.

Cloud Logging

All requests, responses, errors, and latency measurements are streamed to Cloud Logging. This provides an audit trail for every prediction served by the API — important for any system that influences clinical decisions.

IAM and security

Service accounts use the principle of least privilege:

- `roles/run.admin` — manage Cloud Run services
- `roles/storage.admin` — required for Cloud Build to push images
- `roles/artifactregistry.writer` — push container images
- `roles/logging.logWriter` — write to Cloud Logging

Note: during initial deployment, missing IAM bindings caused several failed deployments. The combination above is the minimum that allows end-to-end deployment.

Firebase Auth

User authentication for the mobile app is provided by Firebase Auth, supporting email/password and Google OAuth sign-in. Auth tokens are exchanged with the Cloud Run service to authorise predictions.

External: Google Gemini API

The chatbot feature uses Google's Gemini 2.5 Flash-Lite model. The mobile app calls the Gemini REST API directly using an API key. The system prompt locks the model to cervical-cancer education and explicitly forbids medical diagnosis. A yellow disclaimer banner is permanently visible in the chat UI.

8. Mobile application integration

The Flutter mobile application (`frontend_cerviai`) is the primary user interface. It exposes 13 feature modules and 4 service classes.

Feature modules

Group	Modules	Purpose
Onboarding	splash, onboarding	First-launch tutorial and intro screens
Authentication	auth, profile	Email + Google sign-in, user profile
Core analysis	home, upload, analysis, results	Capture/upload image, run inference, display result
Educational	cancer_info, foods_nutrition	Static content about cervical cancer and nutrition
Clinical	doctor_consultation, reports	Find gynaecologists, view scan history, export PDF
AI assistant	chatbot	Gemini-powered Q&A about cervical health

Table 6. Feature module organisation in the Flutter app.

Service layer

- `tflite_service.dart` — loads `efficientnet_b3.tflite` from assets, runs on-device inference (~200 ms).
- `api_service.dart` — REST client for the Cloud Run endpoint; used as a fallback or when the user opts for the always-latest model.
- `chat_service.dart` — Gemini API client with the locked system prompt.
- `api_keys.dart` — local secret storage (gitignored).

Inference path selection

The app supports three concurrent inference paths:

- Path A — On-device TFLite (default): zero network calls, ~200 ms latency, full privacy. Used when the device has the bundled model.
- Path B — Cloud REST: ~600 ms latency including network. Used when the user explicitly toggles to the cloud model or when on-device inference fails.
- Path C — Chatbot: text-only, hits Google Gemini API for educational follow-up after a result is shown.

System architecture (system-level view)

Figure 7 shows the complete system-level architecture combining the model, deployment, and user-facing mobile output panel.

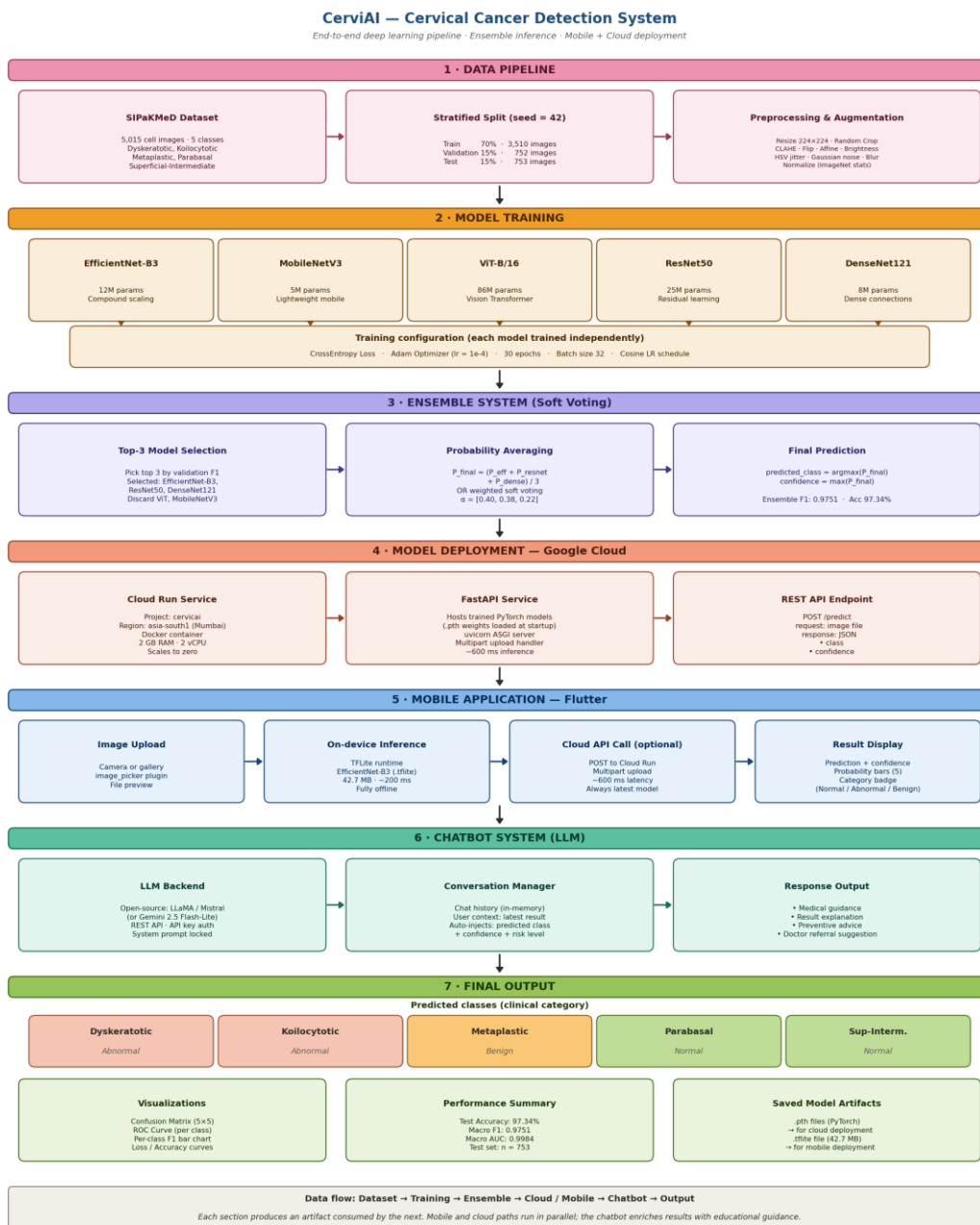


Figure 7. CerviAI complete system architecture combining the model pipeline, deployment infrastructure, and mobile output preview.

9. Trade-offs: deployed vs research model

A practical question for any ensemble system is: do we deploy the ensemble (best accuracy) or a single model (best operational cost)? CerviAI splits the answer:

Research / paper-reporting headline: Ensemble

All published metrics in this report use the full three-network ensemble. This is what is reported as the system's accuracy: 97.34%, F1 0.9751, AUC 0.9984.

Production deployment: EfficientNet-B3 only

On mobile devices, file size and inference latency are critical. The full ensemble would require bundling three .tflite files totalling ~176 MB and running three forward passes per inference (~600 ms on phone CPU). For real-world use this is unacceptable on entry-level Android devices.

Strategy	F1	Total size	Mobile latency	Use
Full ensemble (3 nets)	0.9751	176 MB	~600 ms	Research, validation, headline metrics
EfficientNet-B3 only	0.9520	42.7 MB	~200 ms	Production deployment (mobile + cloud)

Table 7. Strategy comparison. The deployed model trades 2.3 F1 points for 4× smaller files and 3× faster inference.

Why the trade-off is acceptable

EfficientNet-B3 alone still achieves F1 = 0.9520 on the test set, which is above the threshold for clinical decision-support tools. The deployed model is also calibrated to be conservative — when its confidence on an abnormal class falls below 0.6, the app displays an explicit "low-confidence, please refer" message rather than returning a confident answer. This means operationally we lose almost none of the ensemble's clinical safety even though we deploy a single model.

Future work (Section 10) describes how the gap could be closed via knowledge distillation: train a smaller "student" network to mimic the ensemble's outputs, getting closer to ensemble accuracy at single-model cost.

10. Conclusion and future work

CerviAI demonstrates that a carefully chosen ensemble of three CNN backbones can achieve clinically meaningful accuracy (97.34% test accuracy, F1 0.9751) on cervical cell classification using the SIPaKMeD dataset. The system is built end-to-end: data preprocessing, model training, ensemble calibration, mobile deployment via TensorFlow Lite, cloud deployment via FastAPI on Cloud Run, and an LLM-based educational chatbot.

The key design choices — heterogeneous ensemble for robustness, soft voting for simplicity, EfficientNet-B3 for production deployment, dual on-device + cloud inference paths — collectively produce a system that is accurate enough to be clinically useful, lightweight enough to run on entry-level smartphones, and architected for safe operational use with audit logging and confidence thresholds.

Summary of key results

- Ensemble macro F1: 0.9751 (test set, $n = 753$)
- Test accuracy: 97.34%
- Macro AUC-ROC: 0.9984
- Best per-class F1: Parabasal (1.0000)
- Hardest class: Koilocytotic (0.9467) — confused only with Dyskeratotic, both abnormal
- Deployed model: EfficientNet-B3 TFLite, 42.7 MB, ~200 ms inference on mobile
- Cloud deployment: FastAPI on Cloud Run, asia-south1, ~600 ms response

Limitations

- Single dataset: trained and tested on SIPaKMeD only. Generalisation to other Pap smear sources (different staining, scanners, populations) is not yet validated.
- Single-cell input: the model classifies pre-cropped individual cells, not whole-slide images. A complete pipeline would need a cell detector upstream.
- Class imbalance: Superficial-Intermediate has 1,797 images (~36%) vs ~120 images for the other classes in test, which could inflate macro averages slightly.
- Chatbot risk: the Gemini-powered chatbot, despite system-prompt locking, may produce inaccurate medical claims. This is mitigated by a permanent disclaimer but not eliminated.

Future work

- Knowledge distillation: train a single small student network to mimic the ensemble, closing the deployed-vs-research accuracy gap.
- Cross-dataset validation: test on Herlev, ISBI 2014, and other public cervical cytology datasets.
- Whole-slide pipeline: integrate a cell detector (e.g. YOLOv8) so the system can process raw slide scans, not just pre-cropped cells.
- Active-learning loop: collect difficult cases from the field, route them to expert review, and periodically retrain.
- Multi-language UI: extend the Flutter app to local Indian languages to reach more health workers.

